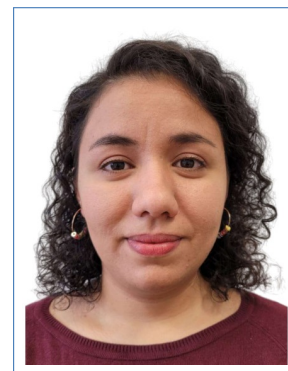


Dr.-Ing. Sarah Ouerghemmi

Curriculum Vitae



Profile

RF and embedded hardware engineer (Dr.-Ing., highest distinction) with 5+ years developing ultra-low-power radio front ends, from concept to validated prototype. Works in interdisciplinary teams on publicly funded consortium projects (BMBF/VDI, DAAD), supervising engineering students and contributing to successful funding proposals. Now looking for an industrial R&D or project leadership role.

Personal information

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Nationality Tunisian

Professional experience

- 03/2021 – Present **Research Engineer — Wake-Up Receiver Development, University of Applied Sciences (HTWK Leipzig)**, Leipzig, Germany
Development and validation of ultra-low-power RF front ends for wireless sensor networks within a research team, from circuit concept through PCB prototype to bench characterization.
- Contributed a wake-up receiver front end operating below 10 μ W while meeting sensitivity targets, enabling continuous connectivity in battery-powered sensor networks
 - Covered the full hardware cycle: 868 MHz LNA and matching network design, ADS simulation, schematic and PCB layout in Altium Designer / Fusion 360, prototype build and MSP430-based bench characterization
 - Supervised final-year engineering students: task planning, weekly project reviews, laboratory resource and component management
 - Supported funding acquisition: co-prepared DAAD proposals (Ta'Ziz, SDG initiatives) and represented the institution at international conferences
- 2022 – 2026 **RF Front-End Development — FUBE Project, HTWK Leipzig, funded by BMBF/VDI**, Leipzig, Germany
Responsible for the wake-up receiver RF front end as part of a multi-partner consortium developing lightweight radio sensors for space-based animal tracking.
- Designed ultra-low-power LNA and matching networks for deep-signal detection, optimizing sensitivity for long-range space-to-sensor links
 - Ran RF simulation in ADS and PCB prototyping in Altium Designer / Fusion 360 through to laboratory validation
 - Provided the measurement basis used by the team for Doppler-based single-satellite localization
- 2022 – 2025 **Partnership & Outreach Coordination — DAAD SDG Project CONCENTRATE, HTWK Leipzig & ENET'COM (Sfax)**, Germany / Tunisia
- Co-organized events and outreach between two partner institutions across two countries
 - Promoted the project at international conferences and supported career centre activities and workshops

Earlier experience & internships

- 09 – 12/2020 **Graduation Project, HTWK Leipzig**, Germany — 868 MHz LNA architecture for wake-up receivers: transistor characterization, ADS simulation and PCB implementation in Altium Designer.
- 07 – 08/2019 **Summer Internship, Sagemcom Software & Technologies**, Tunisia — test, automation and validation team: Azure Cloud gateway testing, Auto Configuration Server (ACS) setup, TR-069, TCP/IP, Linux scripting and Jenkins automation.
- 01 – 03/2019 **End-of-Year Project, PIVA Software**, Tunisia — staff detection and recognition prototype using computer vision and deep learning (Python, OpenCV, TensorFlow/Keras, YOLO, Raspberry Pi).
- 07/2018 **Initiation Internship, Sagemcom**, Tunisia — production of Linky electric meters, from SMD/THD PCB assembly to functionality testing and CPL integration.

Education

- 2021 – 2025 **Doctoral Degree (Dr.-Ing.) — Embedded & Communication Systems, HTWK Leipzig, Germany & ENET'COM**, Sfax, Tunisia
Joint co-supervision between two universities; low-power RF for wireless sensor networks. **Highest distinction with jury congratulations.**
- 2017 – 2020 **Engineering Degree in Electronics & Communication, ENET'COM**, Sfax, Tunisia
Embedded Systems, Communication Networks and Signal Processing. **Ranked 2/60.**
- 2014 – 2017 **Preparatory Cycle Degree (Chemistry & Physics), Preparatory Institute for Engineering Studies of El Manar**, Tunis, Tunisia

Skills overview

Leadership & delivery	Student and junior engineer supervision, project planning and weekly reviews, laboratory resource and component management, funding acquisition (DAAD proposals), international partner coordination, intercultural communication
RF design	Ultra-low-power LNA and matching network design, noise figure and sensitivity optimization, ADS (Advanced Design System); full cycle simulation → schematic → layout → prototype → validation
PCB & schematic	Altium Designer, Fusion 360 Electronics, mixed-signal layout
Lab & test	VNA, spectrum analyzer, oscilloscope, logic analyzer, JTAG/SWD debugging
Embedded systems	MSP430, STM32 (ARM Cortex-M), DSP, Raspberry Pi, FPGA; bare-metal firmware, low-power design, interrupt handling, peripheral drivers (HAL)
Protocols	UART, SPI, I ² C, CAN, TCP/IP, Wireless Sensor Networks (WSN)
Programming	C (advanced), C++, Python, Assembly, MATLAB, VHDL/VHDL-AMS
Methods & tools	Git/GitHub, Agile, Scrum, UML, Linux
Languages	Arabic (native), English (C1), French (C1), German (B1)

Certifications, awards & scholarships

- Linux Foundation **LFS207 — Linux System Administration Essentials**; recipient of the **LiFT Training Scholarship — Women in Open Source**
- 2020 **DAAD Scholarship** — Graduation Project funding at **HTWK Leipzig**
- 2019 **IBM Artificial Intelligence Analyst** — Mastery Award | ERASMUS Soliya Virtual English Exchange Program

Extracurricular activities

- Sports **ENET'COM Handball Team Leader** (2018–2020)
- Club/Event management **Communication Manager — Competitive Programming Club** (2018–2019)
Organization Team Manager — ACM Tunisian Collegiate Programming Contest (2019)
- Competitions **Participant — Tunisian Olympiad of Problem Solving (TOPS)**